

ASSIGNMENT-8.5

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Batch: 36

Task Description #1 (Username Validator – Apply AI in Authentication Context)

- Task: Use AI to generate at least 3 assert test cases for a function `is_valid_username(username)` and then implement the function using Test-Driven Development principles.
- Requirements:
 - o Username length must be between 5 and 15 characters.
 - o Must contain only alphabets and digits.
 - o Must not start with a digit.
 - o No spaces allowed. Example Assert

Test Cases:

```
assert is_valid_username("User123") == True
```

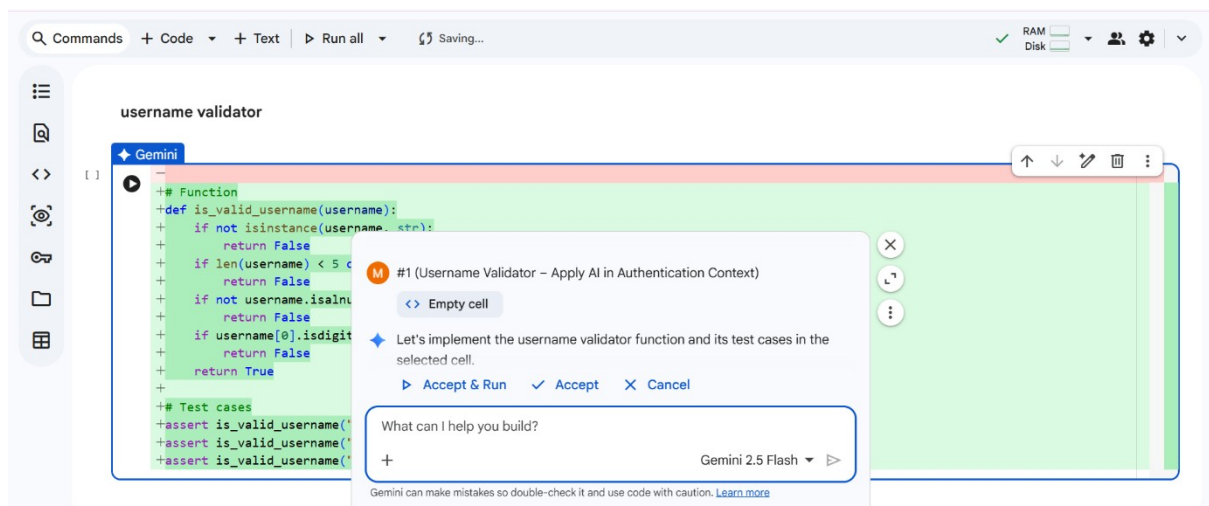
```
assert is_valid_username("12User") == False
```

```
assert is_valid_username("Us er") == False
```

Expected Output #1:

- Username validation logic successfully passing all AI-generated test cases.

Output:



Task Description #2 (Even–Odd & Type Classification – Apply AI for Robust Input Handling)

- Task: Use AI to generate at least 3 assert test cases for a function `classify_value(x)` and implement it using conditional logic and loops.
- Requirements:
 - o If input is an integer, classify as "Even" or "Odd".
 - o If input is 0, return "Zero".
 - o If input is non-numeric, return "Invalid Input".

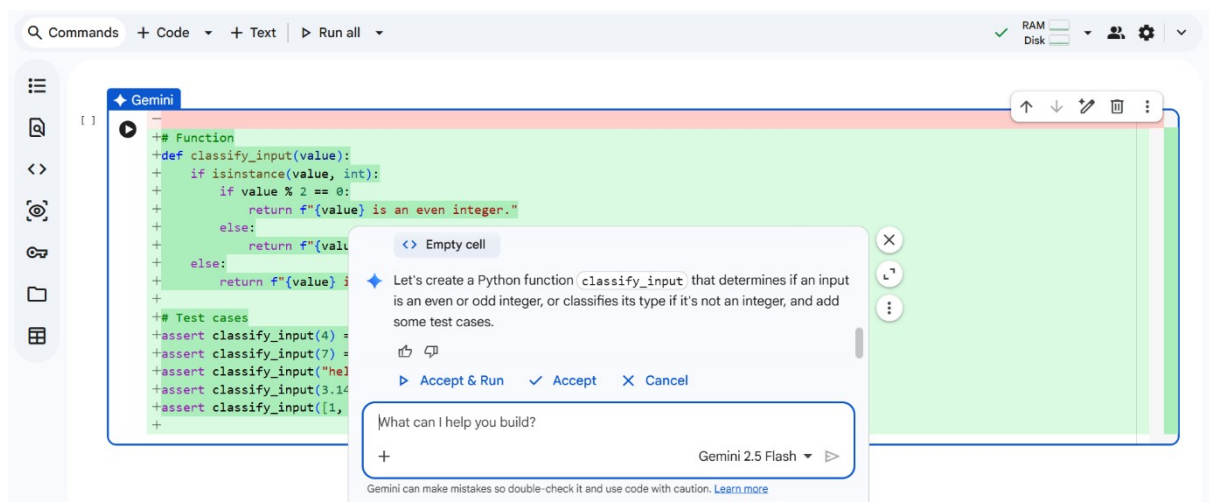
Example Assert Test Cases:

```
assert classify_value(8) == "Even" assert
classify_value(7) == "Odd" assert
classify_value("abc") == "Invalid Input"
```

Expected Output #2:

- Function correctly classifying values and passing all test cases.

Output:



Task Description #3 (Palindrome Checker – Apply AI for String Normalization)

- Task: Use AI to generate at least 3 assert test cases for a function `is_palindrome(text)` and implement the function.
- Requirements:
 - o Ignore case, spaces, and punctuation.
 - o Handle edge cases such as empty strings and single characters.

Example Assert Test Cases:

```
assert is_palindrome("Madam") == True
```

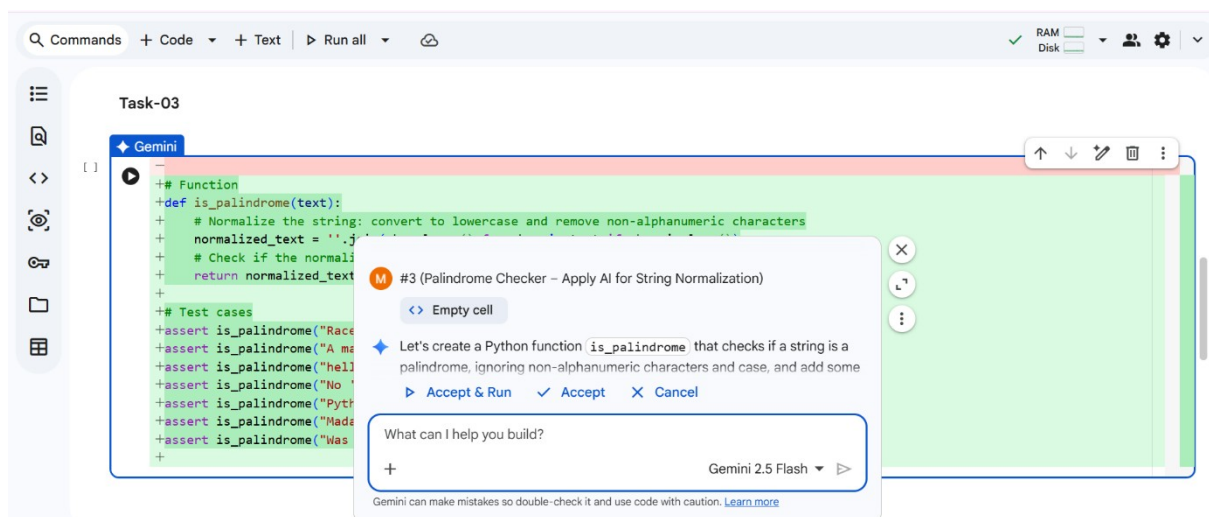
```
assert is_palindrome("A man a plan a canal Panama") ==  
True
```

```
assert is_palindrome("Python") == False
```

Expected Output #3:

- Function correctly identifying palindromes and passing all AI-generated tests.

Output:



Task Description #4 (BankAccount Class – Apply AI for Object-Oriented Test-Driven Development)

- Task: Ask AI to generate at least 3 assert-based test cases for a BankAccount class and then implement the class.
- Methods:

o deposit(amount) o
withdraw(amount)
o get_balance()

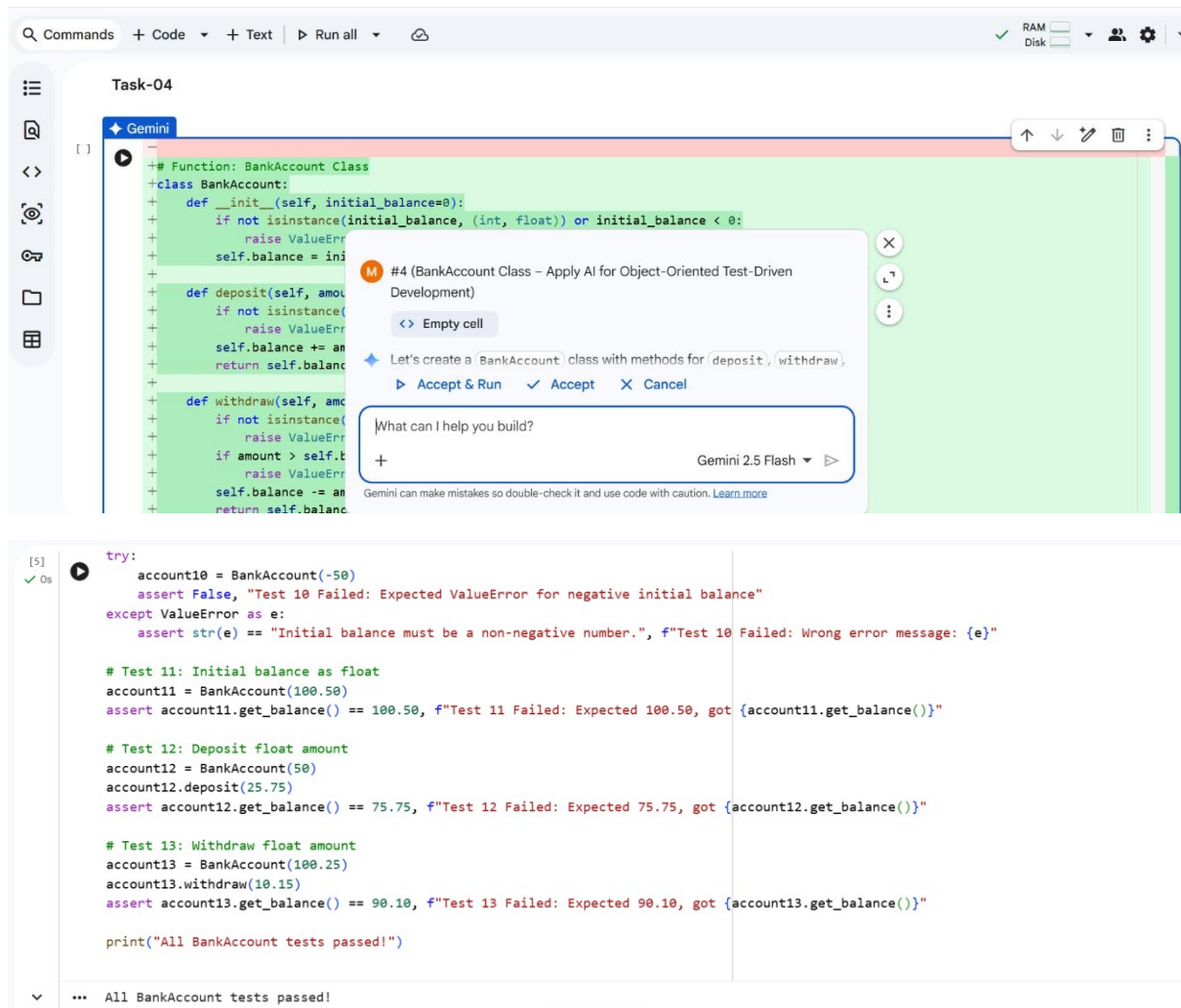
Example Assert Test Cases:

```
acc = BankAccount(1000)
acc.deposit(500)
assert acc.get_balance() == 1500
acc.withdraw(300)
assert acc.get_balance() == 1200
```

Expected Output #4:

- Fully functional class that passes all AI-generated assertions.

Output:



Task Description #5 (Email ID Validation – Apply AI for Data Validation)

- Task: Use AI to generate at least 3 assert test cases for a function `validate_email(email)` and implement the function.
- Requirements:
 - o Must contain `@` and `.`
 - o Must not start or end with special characters.
 - o Should handle invalid formats gracefully.

Example Assert Test Cases:

```
assert validate_email("user@example.com") == True
```

```
assert validate_email("userexample.com") == False
```

```
assert validate_email("@gmail.com") == False
```

Expected Output #5:

- Email validation function passing all AI-generated test cases and handling edge cases correctly.

Output:

